INCH-POUND
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PERFORMANCE SPECIFICATION

SOCKETS, PLUG-IN ELECTRONIC COMPONENTS, ROUND STYLE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers round, plug-in electronic component sockets; for use on panel boards, printed circuit boards, and microelectronic components (see 6.1).

1.2 Classification. Sockets covered by this specification are of the following types as specified (see 3.1).

a. Type II - Printed circuit terminals (see figure 1).

b. Type III - Wire turret terminals (see figure 1).

c. Type IV - Solder-cup terminals (see figure 1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, 3990 East Broad Street, Columbus, Ohio 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-N-290 - Nickel Plating (Electrodeposited).

STANDARDS

MILITARY

MIL-STD-202 - Tests Methods for Electronic and Electrical Component Parts.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-1344 - Test Methods for Electrical Connectors.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following documents(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the issue listed in the current DoDISS and the supplement thereto, if applicable.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

ANSI/NCSL Z540-1-1994 - Calibration Laboratories and Measuring and Test Equipment, General Requirements

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B16 - Rod, Bar, and Shapes for use in Screw Machines, Free Cutting Brass.
ASTM B139 - Rod, Phosphor Bronze, Bar, and Shapes.
ASTM B194 - Plate, Copper Beryllium Alloy, Sheet, Strip, and Rolled Bar.
ASTM B196 - Rod and Bar, Copper Beryllium Alloy.
ASTM D1710 - Standard Specification for Polytetrafluoroethylene (PTFE) Basic Shapes, Rods, and Heavy-Walled Tubing.
ASTM D4066 - Standard Specification for Nylon Injection and Extrusion Materials (PA).
ASTM D5948 - Standard Specification for Molding Compounds, Thermosetting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COPPER DEVELOPMENT ASSOCIATION, INC.

Copper alloy No. CA725
Copper alloy No. CA770

(Application for copies should be addressed to the Copper Development Association, Inc., 405 Lexington Avenue, New York, NY 10017.)

INTERNATIONAL ORGINIZATION FOR STANDARDIZATION

ISO-100012 - Equipment, Metrological Confirmation System for Measuring.

(Application for copies should be addressed to the International Organization for Standardization, 1 Rue De Varembe, Case Postale 56, CH-1211 Geneva 20, Switzerland.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Sockets furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.4 and 6.3).

3.3 Materials. Example reference materials are identified herein. However, when an example reference material is not identified, a material shall be used which will enable the sockets and accessories to meet the performance requirements of this specification. Acceptance or approval of a constituent material shall not be construed as a guaranty of acceptance of the finished product.

3.3.1 Reference critical interface, materials, plating, and processes. The identified reference material, plating, and processes have been established to provide assurances that connectors manufactured to this specification will properly interface to similar industry standard or government specified connector systems without problems of electrochemical contamination of critical electrical or mechanical interfaces or generation of incompatible mechanical interface surface wear products. The manufacturer of connectors supplied to this specification are allowed to use alternate recognized industry standard materials, plating, and processes from those identified in paragraph 3.3 of this specification. Alternate materials, plating and processes used must be coordinated with the qualifying activity as part of the qualification process. Use of alternates to those referenced guidance items by the supplier must not result in inferior short or long term performance or reliability of supplied connectors as compared with connectors manufactured using the referenced materials, plating, or processes. Short or long term failures or reliability problems due to use of these alternates shall be the responsibility of the supplier.

3.3.2 Body. The socket insulator body (and removable insulator, when applicable), shall be manufactured with polytetrafluoroethylene per ASTM D1710, type I, grade 1, class D or diallyl phthalate per ASTM D5948, type SDG-F, glass filled polyamide (nylon) in accordance with ASTM D4066 type PA110 and as specified (see 3.1).

3.3.3 Socket spring contact and socket sleeve. The socket spring contact member shall be beryllium copper in accordance ASTM B139, ASTM B194, ASTM B196, or an equivalent industry standard. The socket sleeve shall be brass in accordance with ASTM B16, or an equivalent industry standard.

3.3.3.1 Contact finish. The contact finish shall be gold applied overall or localized. When contacts are provided in strip form, the absence of plating in the area where the contact was removed from the strip is acceptable, provided it is a non-functional area and any corrosion formed as a result of salt spray testing does not creep into the contact engaging area.

3.3.3.1.1 Overall finish. Contacts shall have be overall gold plated over a nickel underplate. Silver shall not be used as an underplate (see 3.3.1).

3.3.3.1.2 Localized finish. Socket contacts shall be overall nickel plated in accordance with QQ-N-290, class 2, .000030 to .000150 inch thick, or an equivalent industry standard. Localized gold plating applied over the nickel by means of selective plating, inlay, welded dot, or other similar localization systems is permitted in lieu of overall plating, provided the conditions specified herein are met.

3.3.3.1.2.1 Contact mating area. The contact mating area shall be gold plated .000050 inch, over nickel .000050 to .000150 thick (see 3.3.3.1.2). Silver shall not be used as an underplate (see 3.3.1).

3.3.3.1.2.2 Termination's. Termination's shall be plated with either gold .00002 inch minimum or tin-lead .001 inch minimum, as specified, over nickel (see 3.3.1). The following conditions shall apply:

a. Tin-lead shall be 50 to 70 percent tin for type II termination.
b. Tin-lead shall be 50 to 95 percent tin for all other termination types.

3.4. Fungus. Materials used in the construction of these sockets shall be fungus inert and shall be capable of meeting the fungus resistant testing of MIL-STD-810, method 508 or an equivalent industry standard.

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3.5 Interface and physical construction. Sockets shall be of the interface and physical construction specified (see 3.1). The socket insulator body, a socket spring contact member, and a socket sleeve. The entry to the socket contact shall be beveled, chamfered, or tapered to facilitate the engagement of the component lead into the socket. The sleeve shall be machined one piece construction. The socket termination shall be part of the sleeve member. Part numbers applicable to tin/lead finished sleeves shall have a temperature rating of -40° C to +105° C. Part numbers applicable to gold finished spring contacts and gold finished sleeves shall have a temperature rating of -55° C to +125° C.

3.5.1 Wire termination's. The form factor and dimensions of wire termination's shall be as specified (see 3.1 and figure 1).

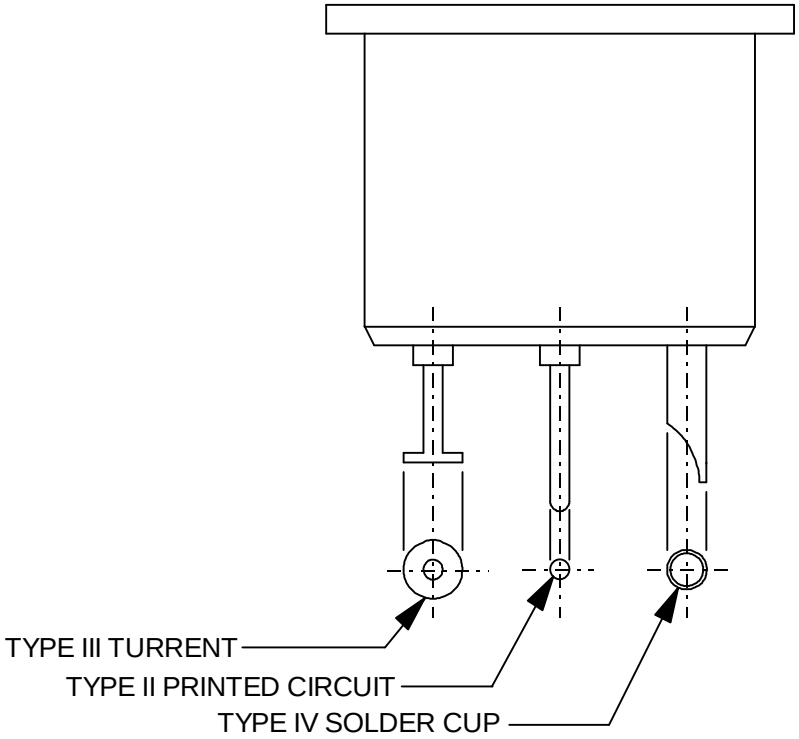


FIGURE 1 Terminal type configurations.

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3.5.2 Body. The body shall be constructed with proper sections and radii so that it will not crack, chip, or break in assembly or in normal service. Depressions, when used to achieve longer creepage paths, shall not cause structural weakness, or moisture entrapment.

3.5.3 Polarization. A polarization feature (mechanical or visual) shall be incorporated in each socket shall be so designed that the mating lead is guided into the contact opening.

3.5.4 Alignment. The socket shall be so designed that the mating lead is guided into the contact opening.

3.5.5 Mounting. When applicable, sockets shall be mounted as specified (see 3.1).

3.5.6 Contact wire wrap post. Unless otherwise specified, (see 3.1), the wire wrap post shall be: 0.0225 inch (0.75 mm)square minimum with a cross-section diagonal 0.0355 inch (0.90 mm) maximum to 0.0325 inch (0.83 mm) minimum. Wire wrap post edge radius shall not exceed 0.003 inch (0.08 mm). Edge burrs shall be 0.0015 inch (0.04 mm), maximum. The wire wrap post shall be parallel within 0.002 inch (0.05 mm) and shall be straight within 0.005 inch (0.13 mm). A minimum wrappable length of 0.515 inch (13.08 mm) shall be maintained after allowing for insertion into a size 2 (0.054-0.070 inch (1.37-1.78 mm)) circuit board, (where applicable). Tip of wire wrap post shall terminate in a radius, bevel, or chamfer to facilitate insertion into the wire wrapping tool.

3.5.7 Mounting standoffs (when specified, see 3.1). Socket bodies intended to be soldered to a printed circuit board shall be provided with mounting bosses so that a minimum of 0.012 inch (0.30 mm) clearance is maintained between the mounting board and the socket body at each terminal location.

3.6 Mating force. When sockets are tested as specified in 4.7.3 the maximum mating force shall be 3/4 pound (3.34 N).

3.7 Contact withdrawal force. When sockets are tested as specified in 4.7.4, the withdrawal force shall be not less than one-half ounce (0.14 N).

3.8 Contact retention. When unmated sockets are tested as specified in 4.7.5, there shall be no damage or loosening of the contacts.

3.9 Insulation resistance. When unmated sockets are tested as specified in 4.7.6, the initial insulation resistance shall be not less than 5,000 megohms.

3.10 Dielectric withstanding voltage. When sockets are tested as specified in 4.7.7, there shall be no evidence of breakdown of insulation or flashover.

3.11 Contact resistance. When sockets are tested as specified in 4.7.8, the initial contact resistance shall not exceed 20 milliohms, and after the durability and corrosive atmosphere tests, the contact resistance shall not exceed the initial value by more than 10 milliohms.

3.12 Capacitance. When sockets are tested as specified in 4.7.9, the capacitance shall not exceed 2 picofarads.

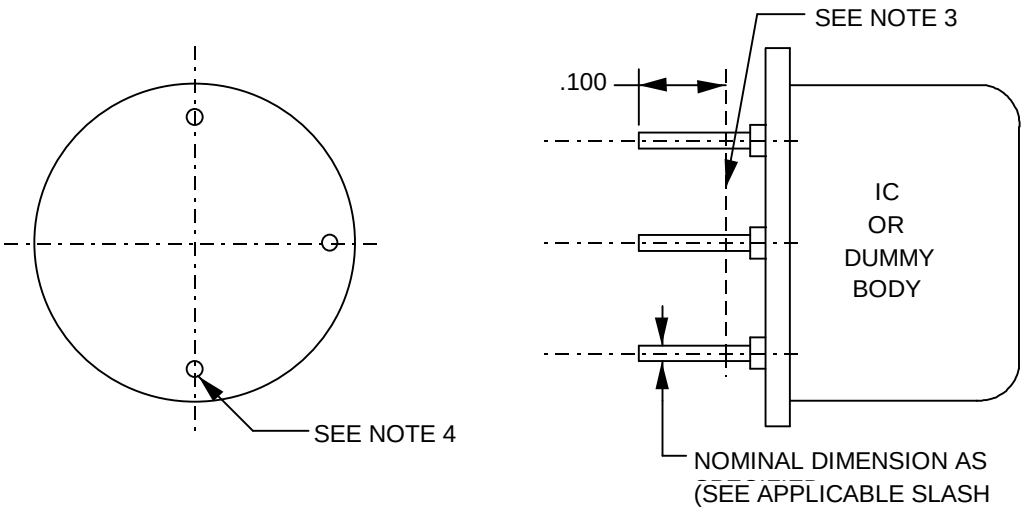
3.13 Vibration. When sockets are tested as specified in 4.7.10 there shall be no physical or mechanical damage to the socket body contacts. During vibration, there shall be no interruption in continuity greater than 1 microsecond of the test circuit which incorporates mated contacts. After the test, the mounting hardware shall show no signs of loosening, fracture, or other deterioration, and the sockets shall meet the contact resistance and contact withdrawal force requirements of 3.8 and 3.7, respectively.

3.14 Shock (specified pulse). When sockets are tested as specified in 4.7.11, there shall be no physical damage to the socket. During the test there shall be no interruption in continuity greater than 1 microsecond of the test circuit which incorporates mated contacts.

3.15 Durability. When sockets are tested as specified in 4.7.12, there shall be no evidence of cracking or breaking. the sockets shall meet the contact resistance and contact withdrawal force requirements of 3.10 and 3.6, respectively.

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3.16 Thermal shock. When a socket with a suitable test plug (see figure 2) is tested as specified in 4.7.13, there shall be no evidence of cracking or crazing of the body or other physical damage to the socket. The socket component shall be capable of mating and unmating without damage to either component.



NOTES
1. Contacts shall be plated gold or tin/lead the same as the contacts of the socket under test.
2. Modifications to the leads such as attaching wires is allowable as long as the area of the lead engaging the socket is unmodified.
3. Minimum engagement length.
4. For pin configuration see individual specification sheets.

FIGURE 2. Mating test plug.

3.17 Humidity. When sockets are tested as specified in 4.7.14, insulation resistance shall be no less than 300 megohms.

3.18 Low-signal level contact resistance. When tested in accordance with 4.7.15, the socket mated with a suitable test plug (see figure 2) shall show no electrical discontinuity and the contact resistance requirement of 3.11 shall not be exceeded.

3.19 Corrosive atmosphere (see 6.5.1). When sockets are tested as specified in 4.7.16, there shall be no evidence of porous plating or exposure of base metal on the contacting surfaces and the low-level circuit requirement of 3.18 shall not be exceeded.

3.20 Solderability. When sockets are tested as specified in 4.7.17, termination's shall withstand the test without damage.

3.21 Resistance to soldering heat (except type I terminals). When sockets are tested as specified in 4.7.18, there shall be no damage.

3.22 Marking. Sockets should be marked in accordance with method I of MIL-STD-1285, and include the military part number, the manufacturer's name or code symbol, and the date code, unless otherwise specified (see 3.1).

3.23 Workmanship. Sockets shall be processed in such a manner as to be uniform in quality and shall be free from burrs, crazing, cracks, voids, pimples, chips, blisters, pinholes, sharp cutting edges, and other defects that will adversely affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.1.1 Verification testing. The following identified tests and test methods assure socket integrity within typical operating conditions and applications. Alternate commercial industry standard test methods are allowed, however when an alternate method is used, the alternate method must be coordinated with the qualifying activity prior to performance of the test. The test methods described herein are proven methods and shall be the referee method in cases of dispute.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment (i.e. ANSI/NCSL Z540-1-1994, ISO 10012-1 part 1, or comparable standards) shall be required.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. Qualification inspection (see 4.4).

c. Verification of qualification (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials, as specified by example, in table I, and on the specification sheets (see 3.1), used in fabricating the sockets, are in accordance with the applicable referenced specifications or requirements prior to such fabrication (see 3.5, 4.7.2, and 6.2).

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. Use of alternate materials, plating, and processes (see 3.3.1) shall be identified for inclusion in the product test documentation.

4.5 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-1344 and MIL-STD-202, or an equivalent industry standard.

4.5.1 Sample size.

4.5.1.2 Group submission. For each specification sheet, ten samples shall be selected of the socket having the largest number of contacts, which shall qualify sockets with a lesser number of contacts of the same design, construction and materials. Two samples of each socket with the lesser number of contacts shall be submitted to visual and mechanical inspection, dimensions and dielectric withstanding voltage. Alternate terminal types specified by a single specification sheet may be qualified by submitting two samples of each type of groups I and II of table I. When different termination finishes are being qualified simultaneously sample sockets of each finish shall be selected so that there are equal, groups of each finish.

Where there is a low profile socket at the same design, to the same specification sheet, it may be qualified at the same time by submitting two samples of the largest number of contacts to groups I and II of table I.

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TABLE I Qualification inspection. 1/

Inspection	Requirement paragraph	Method paragraph
Group I (10 sample units)		
Visual and mechanical	3.1, 3.3, 3.5, 3.22, and 3.23	4.7.2
Mating force	3.6	4.7.3
Contact withdrawal force	3.7	4.7.4
Contact retention	3.8	4.7.5
Insulation resistance	3.9	4.7.6
Dielectric withstanding voltage	3.10	4.7.7
Contact resistance	3.11	4.7.8
Capacitance	3.12	4.7.9
Group II (2 sample units)		
Vibration	3.13	4.7.10
Shock (specified pulse)	3.14	4.7.11
Durability	3.15	4.7.12
Insulation resistance	3.9	4.7.6
Group III (2 sample units)		
Thermal shock	3.16	4.7.13
Insulation resistance	3.9	4.7.6
Humidity	3.17	4.7.14
Dielectric withstanding voltage	3.10	4.7.7
Group IV (2 sample units)		
Low-signal level contact resistance	3.18	4.7.15
Corrosive atmosphere	3.19	4.7.16
Group V (2 sample units)		
Solderability	3.20	4.7.17
Visual and mechanical inspection	3.1, 3.3, 3.5, 3.22, and 3.23	4.7.2
Group VI (2 sample units)		
Resistance to soldering heat	3.21	4.7.18
Visual and mechanical inspection	3.1, 3.3, 3.5, 3.22, and 3.23	4.7.2

1/ sockets with inserted terminals shall not be considered the same design as those with the terminals molded in place.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown. all sample units shall be subjected to the inspections of group I. The sample shall be divided into five groups of two units each and subjected to the inspections for their particular group.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.5.4 Retention of qualification. To retain qualification, the contractor shall verify in coordination with the qualifying activity the capability of manufacturing products which meet the performance requirements of this specification. Refer to the qualifying activity for the guidelines necessary to retain qualification to this particular specification. The contractor shall immediately notify the qualifying activity at any time that the inspection data indicates failure of the qualified product to meet the performance requirements of this specification.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection. Except as specified in 4.6.2.1.2, delivery of products which have passed the group A inspection shall not be delayed pending the results of group B inspection.

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4.6.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all sockets of the same configuration (other than number of pins) produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II, on the same set of sample units in the order shown.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Method paragraph
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.22, and 3.23	4.6.1
Mating force ----------	3.6	4.7.3
Contact withdrawal force -------	3.7	4.7.4

4.6.1.2.1 Sampling plan (group A). Table II tests shall be performed on a production run basis. Samples shall be selected in accordance with table III. If one or more defects are found, the lot shall be screened for that particular defect and defects removed. A new sample of parts shall be selected in accordance with table III and all group A tests again performed. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE III. Lot and sample size.

Lot size	Sample size
1 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10000	29
10001 to	35

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using the next higher lot size specified in table III and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.6.1.2.3 Disposition of sample units. Sample units which have passed all the group A inspections may be delivered on the contract if the lot is accepted and the sample units are still within the specification tolerances.

4.6.2 Periodic inspection. Periodic inspection shall consist of group B. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.2), delivery of products which have passed group A inspection shall not be delayed pending the results of these periodic inspections.

4.6.2.1 Group B inspection. Group B inspection shall consist of the tests specified in table I, in the order shown. Group B inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

4.6.2.1.1 Sampling plan. Ten sample sockets of the largest size for which the manufacturer is qualified and which have been produced on a continuous basis shall be selected at random every 36 months. The sample units shall be divided into five groups of two units each and subjected to the inspections for their particular group.

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4.6.2.1.2 Noncompliance. If a sample fails to pass group B inspection, the contractor shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspections may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action shall be made available to the cognizant inspection activity and to the qualifying activity.

4.7 Methods of inspection.

4.7.1 Test methods. The following identified tests and test method assure socket integrity within typical operating conditions and applications. Alternate commercial industry stand test method are allowed, however when and alternate method is used, the qualifying activity must be notified prior to performance of the test. The test methods described herein are proven methods and shall be the referee method in cases of dispute.

4.7.2 Visual and mechanical inspection. Sockets shall be examined to verify that the dimensions, materials, design construction, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.5, 3.22, and 3.23).

4.7.3 Mating force (see 3.6). The initial force required to fully insert the test gage (see figure 3) shall be measured. The measuring equipment shall conform to the following:

a. The vertical axis of the test gage shall coincide with the vertical axis of the socket.

b. The test gage shall travel along the vertical axis of the socket.

c. The speed of insertion of the test gage into the socket contacts shall not exceed 2 inches per minute for consistent-speed machines, or the rate of loading shall not exceed 80 pounds per minute for contact-rate-of-force machines.

d. Scale mechanism shall have no dashpots or other damping devices.

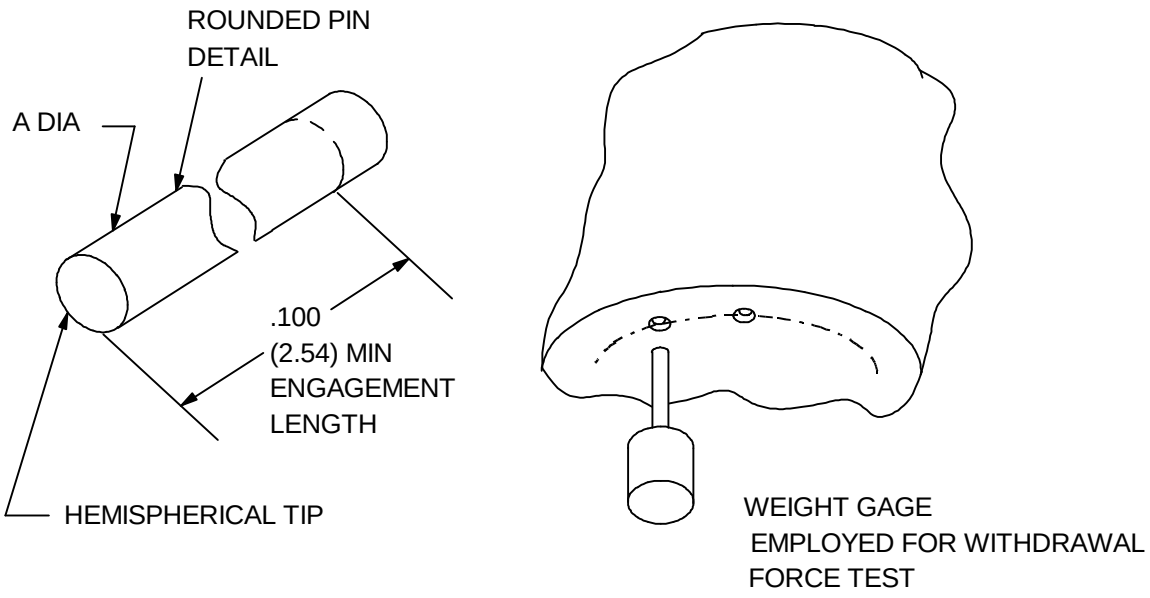
e. Scales shall be calibrated in 1/8 pound steps or less, and shall be accurate to within ∀5 percent.

4.7.4 Contact withdrawal force (see 3.7). After two unmonitored insertions and withdrawals of the maximum mating test gage (see figure 3), the individual withdrawal force shall be measured using the minimum thickness single blade pin (see figure 3). Test pin shall be inserted to a depth of .100 inch (2.54 mm) minimum and the withdrawal force shall be measured.

4.7.5 Contact retention (see 3.8). With the socket mounted to an appropriate fixture, a 7-1/2 pound (3.40 kg) axial load shall be applied to terminals intended for solderless wrap applications, in both directions as that applied by the wrapping tool. a 3/4-pound (0.34 kg) axial load shall be applied in both directions to terminals intended for solder cup applications.

4.7.6 Insulation resistance (see 3.9). Unmated sockets shall be tested in accordance with method 3003 of MIL-STD-1344 , or an equivalent industry standard.

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Contacts to accept mating leads		Max Min Gage	A Dia	
Inch	mm		Inch	mm
.017 +.003 -.001	0.043 +.08 -.03	Max	.0200 +.0000 -.0002	0.501 +.000 -.005
		Min	.0160 +.0002 -.0002	0.406 +.005 -.003
.030 ∀.005	0.76 ∀.13	Max	.0350 +.0000 -.0002	0.889 +.000 -.005
		Min	.250 +.0020 -.0020	.0635 +.051 -.000
.040 ∀.005	1.02 ∀.13	Max	.0450 +.0002 -.0020	1.143 +.005 -.051
		Min	.0350 +.0002 -.0000	0.889 +.005 -.000

NOTES
1. Dimensions are in inches.
2. metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ∀.0005 (.013 mm).
4. Metric equivalents are in parentheses.
5. Material: All pins carbon steel, finish 4 microinch maximum.

FIGURE 3. Insertion and withdrawal gages.

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4.7.7 Dielectric withstanding voltage (see 3.10). Sockets shall be tested in accordance with method 3001 of MIL-STD-1344, or an equivalent industry standard. The following details and exceptions shall apply:

a. Preparation - Mated with suitable test plug (see figure 2) but not mounted on a printed circuit board.

b. Magnitude of test voltage - 600 volts, unless otherwise specified (see 3.1).

c. Nature of potential - AC (rms).

d. Points of application of test voltage - The test voltage shall be applied between all odd-numbered contacts connected together and all even-numbered contacts and all other metallic parts connected together, unless otherwise specified (see 3.1).

e. Method of connection of test voltage to specimen - Affix test probes to termination's described in 4.7.7d, by clips or solder.

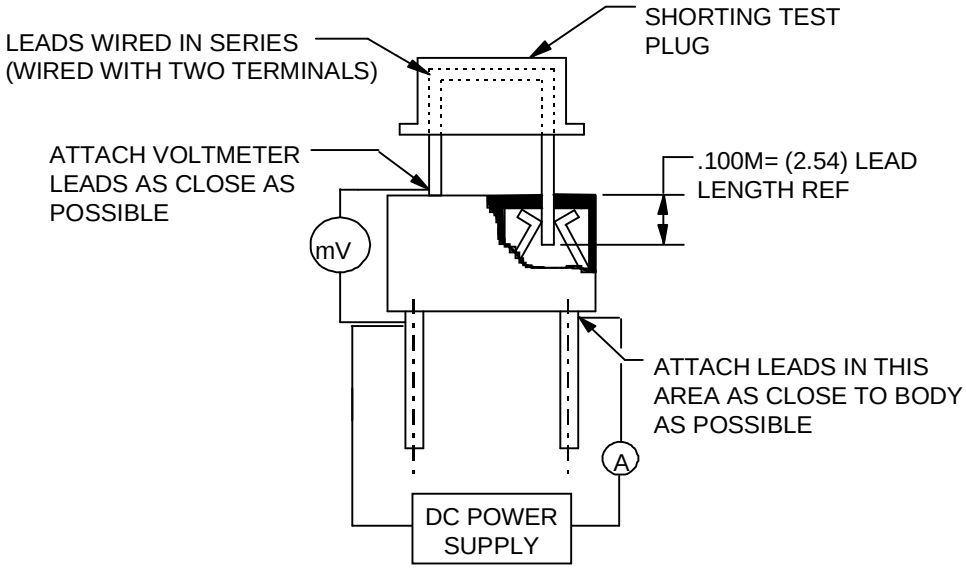
4.7.8 Contact resistance (see 3.11). The contact resistance shall be measured as specified on figure 4. A minimum of three contacts shall be measured on each test specimen in accordance with method 3004 of MIL-STD-1344, or an equivalent industry standard. The following shall apply:

a. Single contact.

b. Brass or copper base alloy gage.

c. Plating to be the same generic materials.

d. Final measurements. Durability (see 4.7.12) and corrosive atmosphere (see 4.7.16) shall be measured followed by contact resistance (see 4.7.7).



NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The measured resistance includes a portion of both the terminal and test socket leads as well as the voltmeter test leads. These resistance's are to be subtracted from the total resistance to obtain the contact resistance.
4. A shorting wire may be soldered between leads in lieu of use of a special shorting plug.
5. Metric equivalents are in parentheses.

FIGURE 4. Contact resistance (low-level and rated current) test setup.

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4.7.9 Capacitance (see 3.12). The socket shall be tested in accordance with method 305 of MIL-STD-202, or an equivalent industry standard. The following details shall apply:

a. Test frequency - 1 kHz.

b. Polarization - Not applicable.

c. Unmounted.

d. Adjacent pins.

e. Seven readings.

4.7.10 Vibration (see 3.13). Sockets shall be tested in accordance with method 2005 of MIL-STD-1344, or an equivalent industry standard. The following details shall apply:

a. Test condition - III.

b. Preparation - Mated with test plugs (see figure 2). (see figure 5 for further details).

c. Following the test, contact resistance and contact withdrawal force shall be measured in accordance with 4.7.8 and 4.7.4, respectively.

4.7.11 Shock (specified pulse) (see 3.14). The socket shall be tested in accordance with method 2004 of MIL-STD-1344, or an equivalent industry standard. The following details shall apply:

a. Mounting method and detection circuit - Mounted by normal means monitoring equipment used shall be capable of detecting circuit interruptions greater than 1 microsecond.

b. Acceleration requirements - Test condition G.

c. Number of blows - One blow in both directions along each of three mutually perpendicular axes for a total of six shocks.

d. Preparation - Mated with a test plug (see figure 2). (see figure 5 for test setup).

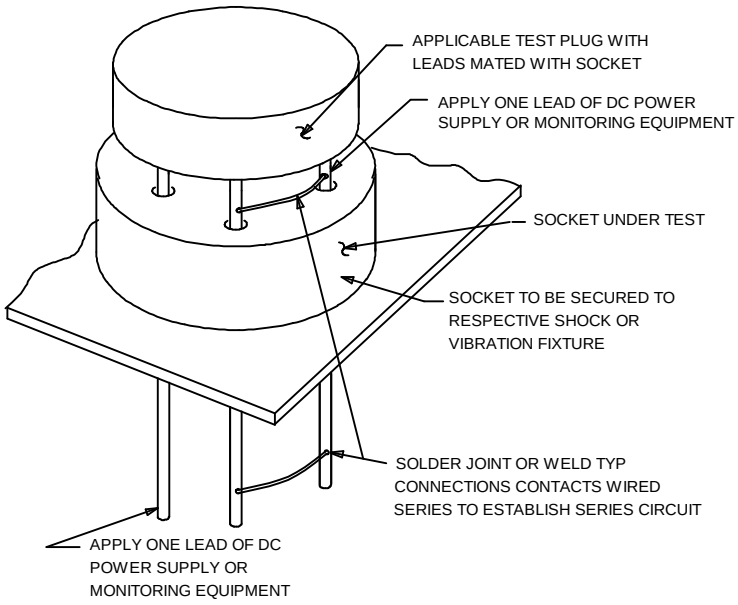


FIGURE 5. Vibration and shock test setup.

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4.7.12 Durability (see 3.15). Each unit shall be subjected to 50 mating and unmating cycles using the test plug (see figure 2). Following the test, contact resistance and contact withdrawal force shall be measured in accordance with 4.7.7 and 4.7.3 respectively.

4.7.13 Thermal shock (see 3.16). sockets shall be tested in accordance with method 1003 of MIL-STD-1344, or an equivalent industry standard. The following details shall apply:

a. Test condition letter - B.

b. Test measurement - The sockets shall be capable of mating and unmating at the temperature extremes (force shall be unmonitored) during the fifth cycle without damage to either component.

4.7.14 Humidity (see 3.17). Sockets shall be tested in accordance with method 1002, type II, MIL-STD-1344, or an equivalent industry standard. Unmounted sockets shall be connected as specified in 4.7.6. Insulation resistance shall be measured in accordance with 4.7.6, within 5 minutes of completion of step 6 of final cycle.

4.7.15 Low-signal level contact resistance (see 3.18). The low-level circuit shall be tested in accordance with MIL-STD-1344, Method 3002, or an equivalent industry standard.

a. Environmental conditioning not required.

b. Seven readings.

c. Final measurements - Contact resistance (see 3.10).

4.7.16 Corrosive atmosphere (see 6.4.1 and 3.11). Mated sockets shall be exposed to a concentrated sulphur atmosphere. The following details shall apply:

a. Sockets that have been mated with a suitable test gage and preconditioned by being subjected to two unmonitored insertions of the insertion force gage (see figure 3) shall be exposed to a 10-25 PPM atmosphere of ammonium polysulfide at a relative humidity of 60 percent or higher at room temperature for 4 hours in an enclosed chamber.

b. At the conclusion of the sulphur atmosphere exposure, the low-level circuit of the undisturbed mated socket shall be measured in accordance with 4.7.15.

4.7.17 Solderability (see 3.20). Each terminal shall be subjected to method 208 of MIL-STD-202, or an equivalent industry standard.

4.7.18 Resistance to soldering heat (see 3.21). Sockets shall be tested in accordance with method 210 of MIL-STD-202, condition B, or an equivalent industry standard. Sockets may be mounted on a glass epoxy wiring board.

5. Packaging.

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military department or Defense Agency, or within the Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. Notes.

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The sockets covered by this specification are intended for use on a printed circuit board or mounting into chassis. Terminals may be soldered to the printed circuit board unless otherwise noted (see 3.1). Only plug-in component leads with similar finishes to the mating socket contacts should be mated, reference MIL-HDBK-454, General Guidelines for Electronic Equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Title, number and date of the applicable specification sheet, and the complete PIN (see 1.2.1 and 3.1).

c. Certificate of compliance covering materials, when required.

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

e. Packaging requirement (see 5.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 83502 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Defense Supply Center Columbus, ATTN: DSCC-VQ, 3990 East Broad Street, Columbus, Ohio 43216-5000. Application procedures should conform to the "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Provisions governing qualification. Copies of "Provisions Governing Qualification" may be obtained upon application to Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.4 Definition.

6.4.1 Corrosive atmosphere. The corrosive atmosphere test is intended to reveal imperfections in the plated contacting surfaces such as pores, scratches, or incomplete plating coverage. It will also reveal defects such as low contact pressure. However, it is not intended to correlate directly with long term atmospheric contamination.

6.5 GUIDANCE INFORMATION

6.5.1 Overall finish. Based on past experience, to meet the performance of this specification, contacts , have been fabricated with overall gold, plated in accordance with MIL-G-45204 type I, grade C, class 0 for the socket spring contact and class 00 for the socket sleeve. Finish was over a nickel underplate in accordance with QQ-N-290, .000030 to .000150 inch thick. Silver was not allowed as an underplate.

6.5.2 Contact mating area. Based on past experience, to meet the performance of this specification, the contact mating area, have been fabricated with gold plating in accordance with MIL-G-45204, type II, grade C, class 0, over nickel. Silver was not allowed to be used as an underplate.

6.5.3 Termination. Based on past experience, to meet the performance of this specification, termination's have been fabricated with either gold per MIL-G-45204, type II, grade C, class 00, or tin-lead per MIL-P-81728, .0001 inch minimum, as specified (see 3.1), over nickel (see 3.3.2.1.2). The following conditions did apply:

a. Tin-lead should be 50 to 70 percent tin for type II termination.

b. Tin-lead should be 50 to 95 percent tin for all other termination types.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.7 Subject term (keyword listing).

Contacts
Current
Dielectric
Finish
Heat
Humidity
Inspection
Materials
Mounting
Mating force
Polarization
Qualification
Sampling
Socket
Solder
Turrent

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AT, AV, MI
Navy - AS, SH, MC
Air Force - 17, 19, 99

Preparing activity:
DLA -CC

(Project 5935-4034)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-83502C	**2. DOCUMENT DATE (YYMMDD)** 970407

3. DOCUMENT TITLE SOCKETS, PLUG-IN ELECTRONIC COMPONENTS, ROUND STYLE, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center Columbus	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0573 (2) DSN 850-0573 (3) Fax 614-692-3969
c. ADDRESS (Include Zip Code) DSCC-VAI 3990 East Broad Street Columbus, Ohio 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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